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Modeling Guidelines for Synthesizable RTL



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KV VLSI Design, 336.051, WiSe 26/27

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Today's Goals

After this lecture, you can

- explain why the **RTL and the netlist** must simulate alike
- predict how simulators treat the **unknown value x**
- find a **missing reset** before the chip does
- replace **FPGA habits** by ASIC practice
- replace a **divided clock** by a **clock enable**
- design **macro boundaries** that are easy to close and verify
- compare **architectures** by area and timing



Chapter 3

Lecture notes: Chapter 3 of iic-jku.github.io/vlsi-design

Simulation Must Match the Netlist

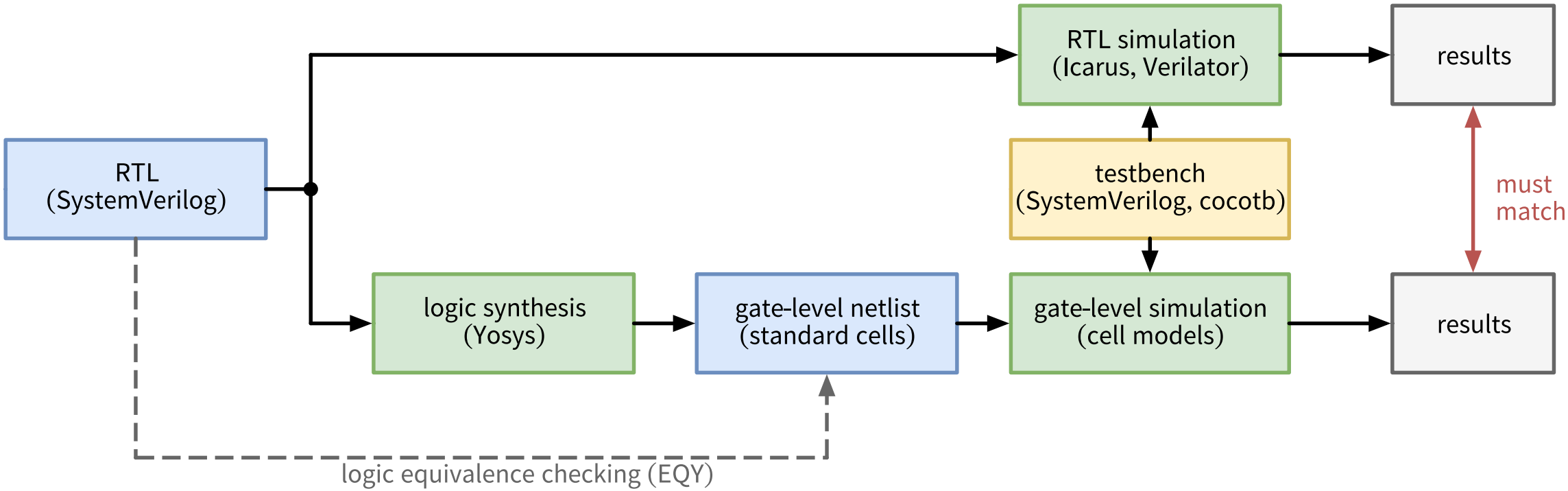


Levels of Abstraction

Level	Describes	Example
behavioral	the function only	Python model, testbench
RTL	registers and the logic between them	your SystemVerilog
gate level	a netlist of standard cells	output of Yosys
layout	polygons on the masks	GDSII

- You write and simulate the **RTL**
- The flow turns it into the **netlist**, the netlist into the **layout**

The Chip Implements the Netlist



- Same testbench on **RTL and netlist**, the results **must match**
- Logic equivalence checking **proves** it without simulation (Chapter 5)

Where Mismatches Come From

- Debugging a netlist is hard: generated names, no FSM states, flat hierarchy
  - template counter, final netlist: the flip-flop outputs are `net3` to `net10`
- **Simulation semantics** that synthesis ignores: `x`, initial values
- **Constructs tools read differently**: sensitivity lists, `full_case`, `unique` without `default` (Chapter 2)
- **Timing** the RTL simulation does not model: glitches, derived clocks
- So: write RTL that **cannot** cause a mismatch

Four-Valued Simulation, Two-Valued Silicon



0, 1, x, z

- `logic` has four values: `0`, `1`, `z` (high impedance), `x` (unknown)
- Silicon has two: every node is **0 or 1**, `x` means the **simulator does not know**
  - register before its first load, two drivers, `'x` assignment, index out of range
- **X-optimism**: a known value where the hardware could have either
  - `if` takes the `else` branch, `case` runs the `default`
- **X-pessimism**: `x` where the hardware has a known value
  - e.g., a `default` that assigns `'x` (next slide)

Experiment: A Multiplexer with sel = x

```
always_comb
  if (sel) y_if = b;
  else     y_if = a;
always_comb
  case (sel)
    1'b0:    y_case = a;
    1'b1:    y_case = b;
    default: y_case = 1'bx;
  endcase
assign y_cond = sel ? b : a;
assign y_and  = sel & a;
```

sel = x	Icarus	Verilator	Hardware
if, else	0 / 1	0 / 1	0 or 1 / 1
case, default x	x / x	0 / 1	0 or 1 / 1
sel ? b : a	x / 1	0 / 1	0 or 1 / 1
cell mux2_1	x / 1	not run	0 or 1 / 1
sel & a	0 / x	0 / 0	0 / 0 or 1

each entry: a = 0, b = 1 / a = b = 1

Testbench: `xprop/testbenches/verilog/xprop_tb.sv`

What the Table Shows

- **`if` is optimistic**: shows `0` where the hardware has `0` or `1`
- **An `'x` default is pessimistic**: `x` although the hardware gives `1`
- **`?:` behaves like the mux2_1 cell**: equal bits stay known, different bits become `x`
- **The AND gate is exact** in Icarus: `x & 0 = 0`
- **Verilator knows no x**: two-state, here every `x` became `0`
  - `+verilator+rand+reset+2 +verilator+seed+<n>`: random start values

Guidelines for x

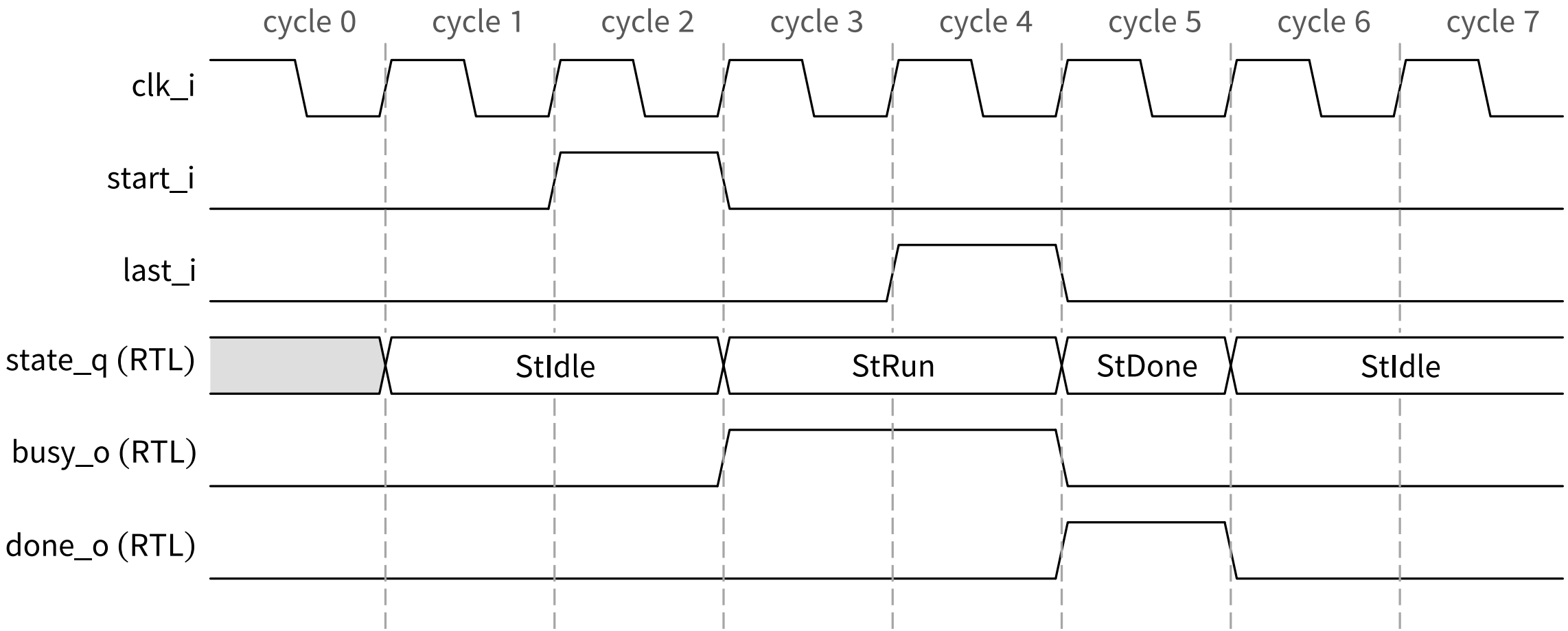
- **Never assign `'x` in RTL**
  - lowRISC: “The use of X literals in RTL code is strongly discouraged”
  - synthesis may pick any value, the netlist differs from the simulation
- **No x in the RTL simulation is no proof**
- Run Verilator with **random start values** and several seeds
- Simulate the **netlist** at least once
- Check with `!==` or `$isunknown()` in testbenches

The Missing Reset

```
// State register without a reset
always_ff @(posedge clk_i) begin
  state_q <= state_d;
end
```

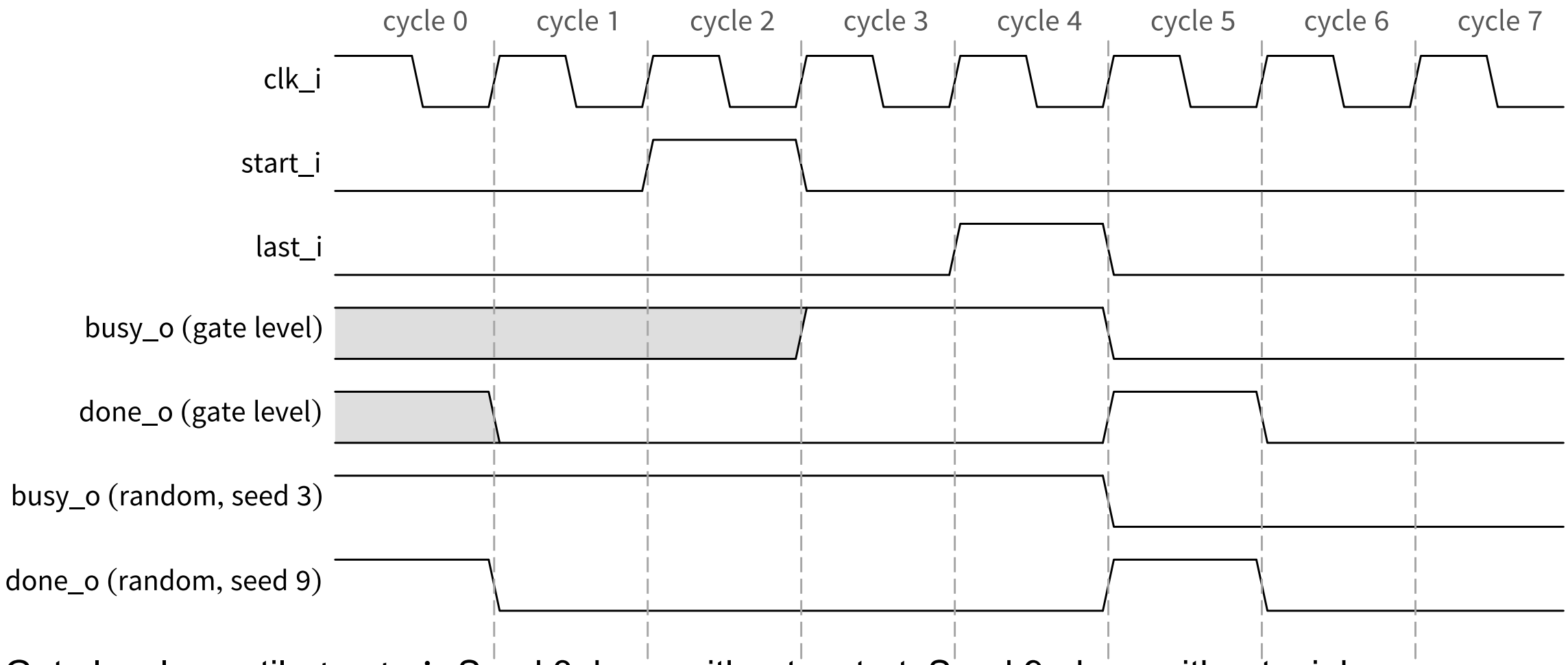
- The controller FSM of Chapter 2, **reset removed**
- `start_i` in cycle 2, `last_i` in cycle 4
- Four simulations of the **same RTL**:
  - Icarus Verilog
  - Verilator
  - Verilator with **random start values**
  - Icarus Verilog, **netlist** on SG13CMOS5L cells

The RTL Simulation Hides It



No item matches `xx`, the `default` branch returns to `StIdle`: the outputs **never show x**

The Netlist and Random Start Values Show It



Gate level: `x` until `start_i`. Seed 3: busy without a start. Seed 9: done without a job.

Live Demo: Four Simulations

```
cd content/modeling/code/ctrl_fsm_noreset
iverilog -g2012 -o rtl.vvp rtl/*.sv testbenches/verilog/*.sv
vvp rtl.vvp                                    # Icarus Verilog
verilator --binary --timing --timescale 1ns/1ps \
  --top-module ctrl_fsm_noreset_tb rtl/*.sv testbenches/verilog/*.sv
./obj_dir/Vctrl_fsm_noreset_tb                 # Verilator
./obj_dir/Vctrl_fsm_noreset_tb +verilator+rand+reset+2 +verilator+seed+3
yosys -p "read_verilog -sv rtl/ctrl_fsm_noreset.sv; synth ...; write_verilog nl.v"
iverilog -g2012 -gno-specify -DGL ... nl.v testbenches/verilog/*.sv
vvp a.out                                      # gate level
```

- Complete commands: exercise in Section 3.3.1

Guidelines for Resets

- **Reset every register with control state**: FSM states, counters, valid flags, pointers
- A **data register** may stay without reset if it is written before it is read
  - saves the reset net, not cell area: every SG13CMOS5L flip-flop has a reset input
- Use the four-state **`logic`**, also as the base of an enumeration
  - `bit` and `int` start at `0` and hide a missing reset, like Verilator
- VHDL: an enumeration starts with its **leftmost value**, the same trap

Exercise (Section 3.3.1): find the missing reset in four simulations.

FPGA Habits That Break on an ASIC



FPGA Habits and Their ASIC Counterparts

FPGA habit	On an ASIC	Instead
initial values	dropped by synthesis	a reset
register output as clock	a second clock and tree	a clock enable
inferred block RAM	flip-flops, 10× the area	an SRAM macro
inferred multiplier (DSP)	a large, slow cell array	a suitable architecture
internal tri-state bus	can short or float	a multiplexer
PLL, clock manager	not in the cell library	analog IP, clock gating
global reset at configuration	no automatic reset	reset pin, synchronizer

Initial Values

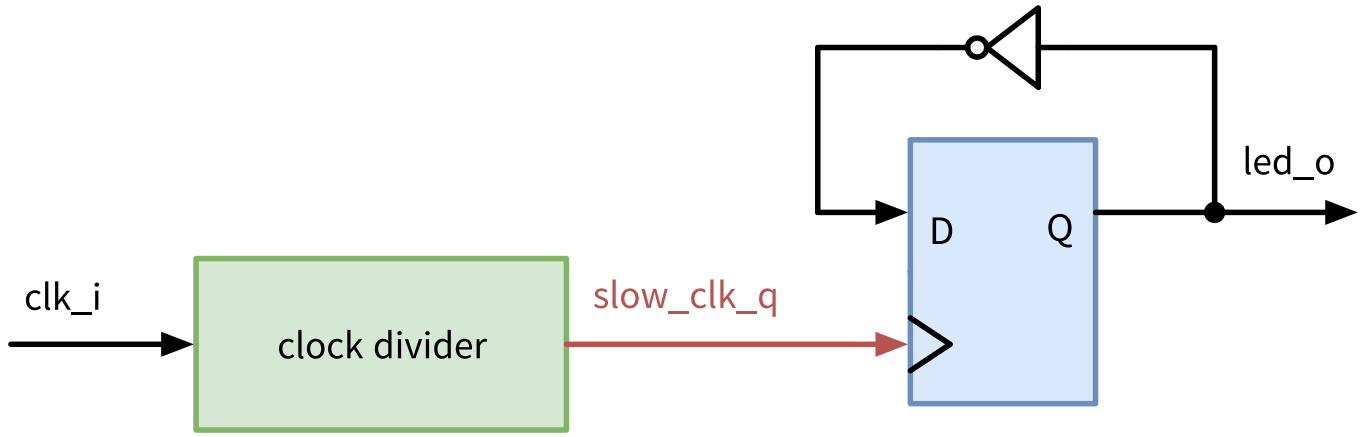
```
logic [COUNT_WIDTH-1:0] count_q    = '0;
logic                   slow_clk_q = 1'b0;
logic                   led_q      = 1'b0;
```

- **RTL simulation** honors them: Icarus and Verilator start at zero
- **Verilator `-Wall`** warns about them, see the exercise
- **Yosys** drops them when mapping onto SG13CMOS5L cells, **without a warning**
- **Netlist** (`DIV = 4`): `led_o` is still x after 40 cycles, the chip starts **random**
- `initial` is for testbenches only, the **reset** sets start values

Why a Divided Clock Hurts

- A second **clock**: every path between the two must be timed
- A second **clock tree**, skewed against `clk_i`
- A **generated clock** in the constraints (Chapter 6)
- The **scan test** must control it (Chapter 9)
- Without reset: `x`, **no edge** at all

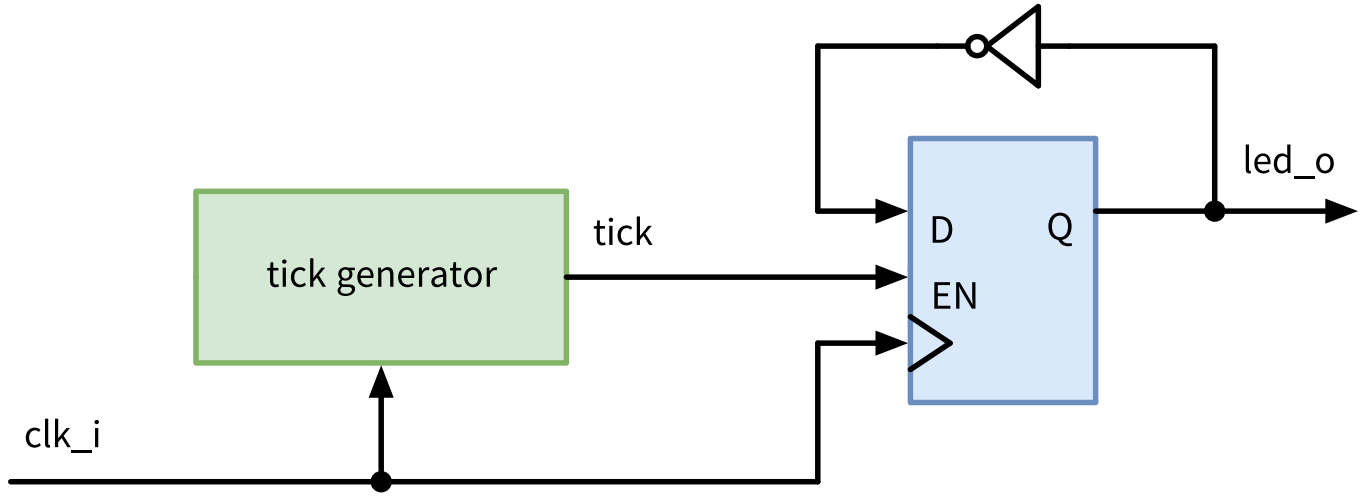
(a) divided clock: two clock domains



The Clock Enable

- Every **rising edge** of the slow clock becomes a **one-cycle `tick`**
- All registers stay on **`clk_i`**
- One clock, one tree, one set of constraints
- Idle blocks: switch the clock off with **clock gating cells** (Chapter 7)

(b) clock enable: one clock



Exercise (Section 3.4.2): an LED blinker from FPGA to ASIC.

Memories: Flip-Flops or an SRAM Macro?

256 words × 8 bits	Cell area	Per bit
flip-flops and multiplexers (Yosys)	165,743 µm²	81 µm²
SRAM macro RM_IHPSG13_1P_256x8_c3_bm_bist	17,547 µm²	8.6 µm²

- Flip-flops: 2056 `dfrbpq_1`, 2074 `mux2_1`, 368 `mux4_1`, **9.4× the area**
- Even 32 × 8 bits (20,944 µm²) exceed the smallest SRAM, 64 × 16 (15,240 µm²)
- **Flip-flops**: small FIFOs, register files with several read ports
- **SRAM macro**: everything larger, a hard macro with model, timing, LEF, GDS

Tri-States

```
tri [7:0] bus;
assign bus = sel_a_i  ? a_i : 8'bz;  // driver A
assign bus = !sel_a_i ? b_i : 8'bz;  // driver B
assign y_o = bus;
```

- Verilator: no warning. Plain Yosys `synth`: **error**, `y_o` “already driven”
- LibreLane maps it onto `sg13cmos5l_ebufn_2` tri-state buffers
- Still avoid it: overlapping enables **short**, no enable **floats**
- lowRISC: “Do not use three-state logic … to accomplish on-chip logic”
- Write a **multiplexer**, tri-states only in the **pads**

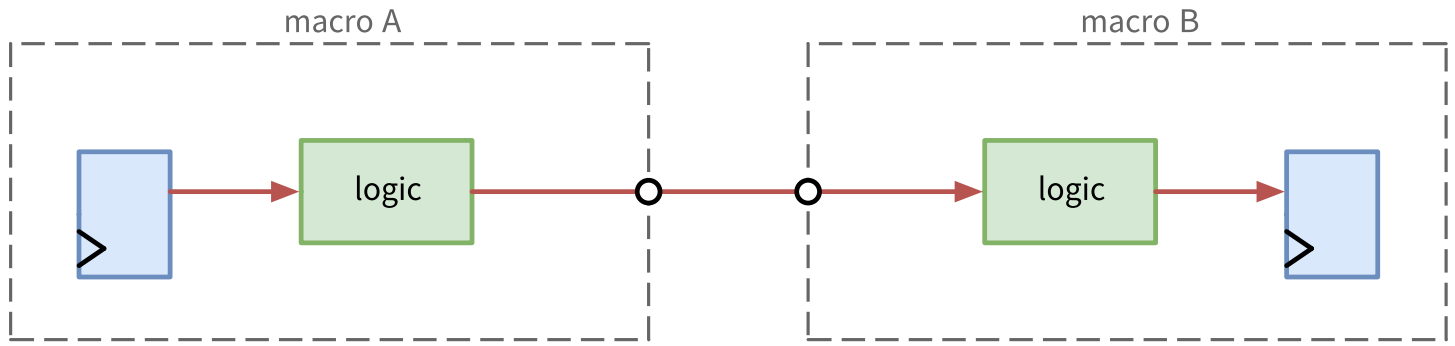
Macro Boundaries



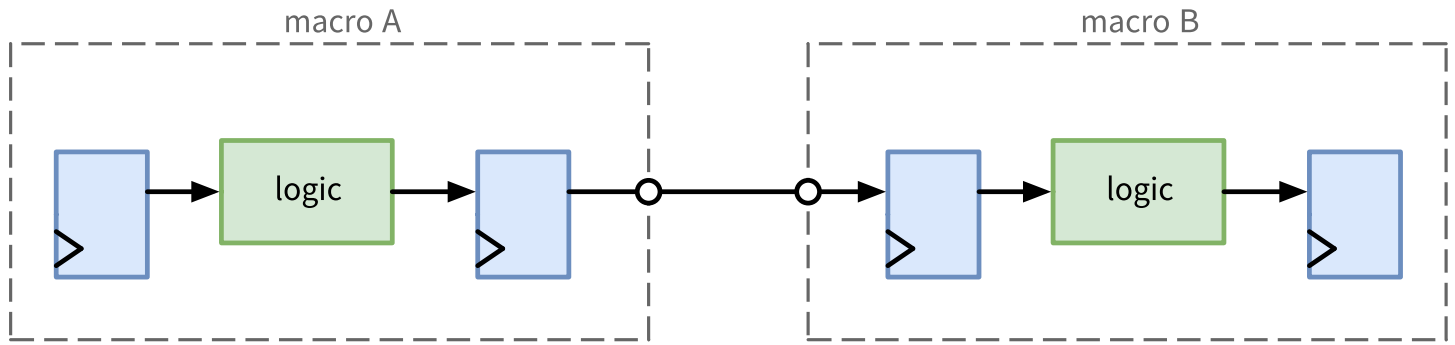
Registered Inputs and Outputs

- Macros are **hardened separately**
- LibreLane assumes **20 %** of the period outside
  - 4 ns of the 20 ns of the counter
- Register every **output**, inputs from far away too
- Price: a cycle of **latency**

(a) unregistered boundary



(b) registered outputs and inputs



Ports That Survive Synthesis

- Same cocotb testbench at RTL and gate level: `make sim-gl-cocotb`
- **Plain `logic` vectors** at the top-level ports
- **No top-level parameters** that the testbench changes
- **Access only the ports**: internal names do not survive
- **A reset** at the start of every test, the netlist starts as `x`

Architecture Determines the Result



Synthesis Keeps Your Architecture

- Synthesis **optimizes the logic** you describe
- It does **not invent** a new architecture
  - a binary counter stays a binary counter, a shift register a shift register
- The **architecture** decides area and speed, much more than the code
- Case study: the **exclusive counter**

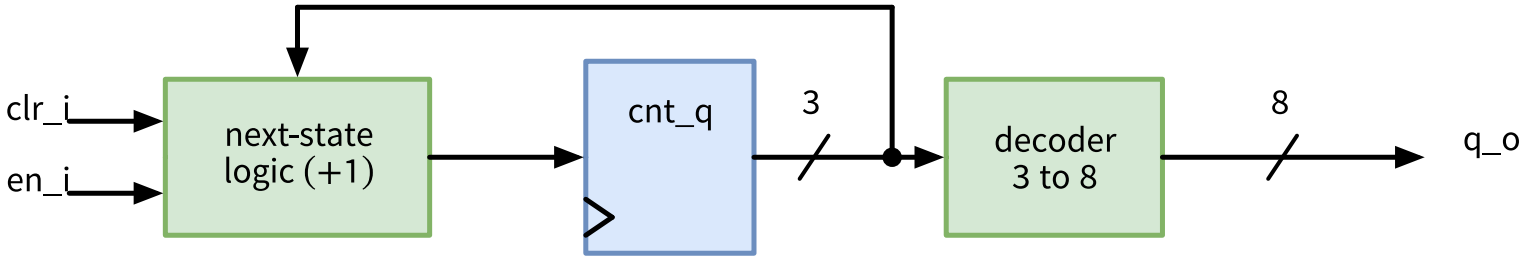
The Exclusive Counter

- Eight outputs `q_o[7:0]`, **exactly one is high** (one-hot)
- Reset `rst_ni` and synchronous `clr_i` (priority over `en_i`): `q_o[0]` high
- `en_i`: the high bit moves up by one, **wraps** from `q_o[7]` to `q_o[0]`
- Otherwise `q_o` keeps its value

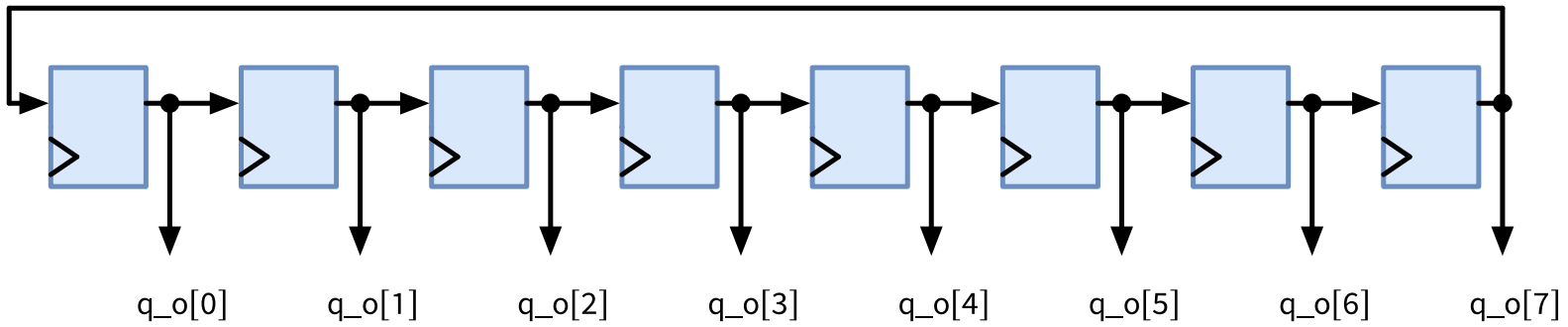
```
q_o: 00000001 -> 00000010 -> 00000100 -> ... -> 10000000 -> 00000001
```

Two Architectures

(a) binary counter and decoder



(b) one-hot ring



Two Descriptions

Binary counter and decoder

```
always_comb begin
  cnt_d = cnt_q;
  if (clr_i) begin
    cnt_d = '0;
  end else if (en_i) begin
    cnt_d = cnt_q + 3'd1;
  end
end
assign q_o = 8'b0000_0001 << cnt_q;
```

One-hot ring

```
always_comb begin
  q_d = q_q;
  if (clr_i) begin
    q_d = 8'b0000_0001;
  end else if (en_i) begin
    q_d = {q_q[6:0], q_q[7]};
  end
end
assign q_o = q_q;
```

- Same register process with `rst_ni`, same cocotb testbench: both pass

Results on SG13CMOS5L

	Binary + decoder	One-hot ring
flip-flops	3	8
standard cells	22	28
cell area	308.4 µm²	553.2 µm²
clock to output	0.49 ns	0.23 ns
flip-flop to flip-flop	0.52 ns	0.31 ns

typical corner, 10 fF load, ideal clock (Yosys, OpenSTA)

- Ring: **+79 % area**
- Ring: **faster**, clean outputs
- Binary: one-hot **by construction**
- `for` loop variant of the ring: **identical** netlist

A Third Architecture

- **Johnson counter**: 4 flip-flops, shift left, feed back the inverted MSB `~s_q[3]`
- Eight states, **one bit changes** per counting step
- Decoder: one **2-input gate** per output
- Where does it rank in area and speed?

Exercise (Section 3.6.1): a third architecture, the Johnson counter.

Checklist and Summary



Checklist for ASIC-Ready RTL

- Reset all **control state**
- No **initial values**, no **'x**
- **One clock**: a clock enable, no derived clocks
- Larger memories as **SRAM macros**
- No internal **tri-states**
- **Registered** macro outputs, **plain** top-level ports
- The **architecture** first, measured with Yosys and OpenSTA
- Simulate in **Icarus**, **Verilator** with random start values, and at **gate level**

Summary

- You verify the RTL, the chip implements the **netlist**: they must agree
- `if` and `case` misjudge `x`, `?:` matches the mux2_1 cell, **Verilator shows no x**
- A **missing reset** hides in RTL: random start values and the netlist reveal it
- **FPGA habits** do not carry over: initial values, divided clocks, inferred RAMs, tri-states
- **Clock enable** instead of a divided clock, **registered** macro outputs
- The **architecture** decides area and speed, not the coding style

Exercises

1. Find the Missing Reset (3.3.1)
2. An LED Blinker from FPGA to ASIC (3.4.2)
3. A Third Architecture, the Johnson Counter (3.6.1)

All code in `content/modeling/code` of the course repository, every number reproduced by `scripts/check_modeling.sh`



Chapter 3



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